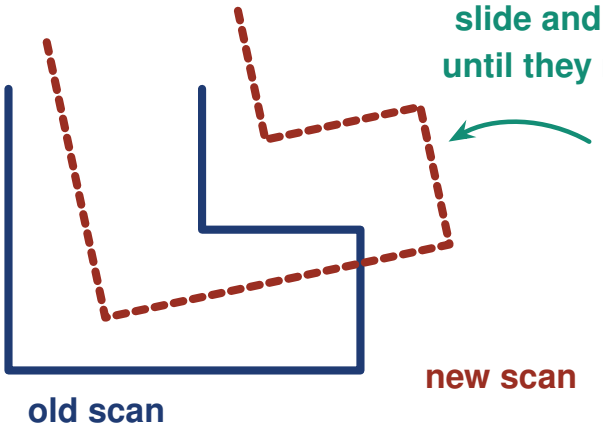
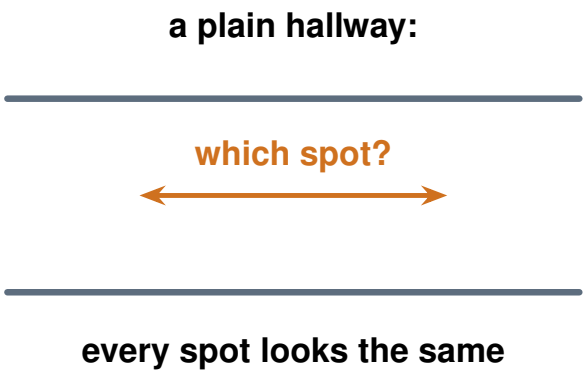
Scan Matching: Fitting Clouds Together

BIG IDEA

Slide and twist the new scan until it lines up with the old one.





THE IDEA

Draw a shape on a see-through sheet. A friend draws the same shape on another sheet, a little shifted and turned. Lay one sheet on the other and slide and twist the top one until the drawings line up. How far you slid and twisted it tells you how your friend's sheet was placed. A LiDAR does this with two scans to learn how far the car moved.

REMEMBER

- ✓ Matching two scans tells how far the scanner moved and turned.
- ✓ Sharp corners and flat walls are the easiest things to match.
- ✓ In a long, plain hallway everything looks alike, so add targets or cross the hallway.

TRY THIS

Trace a shape on two sheets of tracing paper. Shift and turn one, then slide it back until the shapes match.